Quiz: Section 2.1 **Name (Print):** _____
MAT 2000
Introduction to Statistics
Grace College

---

1. An environmental science teacher at a high school with a large population of students was curious whether the majority (more than half) of students at the school regularly recycle plastic bottles. The teacher selected a random sample of 146 students at the school to interview, then asked each student: "Do you recycle plastic bottles more often than you throw them in the trash?" (Yes/No). Suppose that 89 of the students answered "Yes".

   How could coins, a spinner, or cards be used to simulate one of the samples in the simulated null distribution? Select the option that does NOT apply.

   A. Flip a coin 146 times, where heads represents "answered Yes." Count the number of heads in 146 flips.

   B. Create a spinner with 61% shaded to represent "answered Yes." Spin the spinner 146 times and count the number of times the spinner lands in the shaded area.

   C. Have 5 black cards and 5 red cards where black represents "answered Yes." Draw with replacement from the cards 146 times and count the number of times a black card is drawn.

   D. Create a spinner with 50% shaded to represent "answered Yes." Spin the spinner 146 times and count the number of times the spinner lands in the shaded area.

2. A test is being performed to see whether a dog, Hope, understands pointing towards an object. Two objects are placed about 2.5 meters away, then the tester point towards one of the objects. In 20 trials, Hope goes to the correct object 13 times (or 65Finish filling in the blanks with the correct One Proportion applet inputs to carry out an appropriate simulation of this process, if Hope does not understand pointing towards an object and is just guessing.

   - Probability of success: ___(1)____
   - Sample size: ___(2)____
   - Number of samples: Any integer as large or larger than 1000

   A. 0.65, 13 B. 0.65, 20 C. 0.5, 13 D. 0.5, 20

3. Which of the following is NOT a step in the 3S Strategy for measuring strength of evidence?

   A. Strength of evidence B. Statistic C. Sample D. Simulate

4. Do red uniform wearers tend to win more often than those wearing blue uniforms in tae-kwon-do matches where competitors are randomly assigned to wear either a red or blue uniform? In a sample of 80 tae-kwon-do matches, there were 45 matches where the red uniform wearer won.

   What is the value of the standardized statistic?

   A. $z = 0.5625$

   B. $z = -0.5625$

   C. $z = 1.12$

   D. $z = -1.12$